

Observing the Intersections of the Complex Plane's Four Objects

(Exercise XXIX)

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Abstract

We observe, directly and with machine-checked proofs, the intersections of the four central objects of the Riemann direction: the real axis, the imaginary axis, the critical circle, and the prime circles. The complete intersection table is established: the real axis meets the critical circle at $\{0, 2\}$ and the imaginary axis at $\{0\}$; the real axis meets the prime circle of p at $\{\pm\sqrt{p}\}$ and the imaginary axis at $\{\pm i\sqrt{p}\}$; the prime-2 circle meets the critical circle at $\{1 \pm i\}$; the prime-3 circle meets it at $\{3/2 \pm \sqrt{3}/2 \cdot i\}$; and — the new structural theorem — the prime circle of $p \geq 5$ does not meet the critical circle at all: a point of both circles would force $a = p/2$ and $b^2 = p(4 - p)/4 < 0$. Equivalently: the prime circle (radius $\sqrt{p}$, center 0) meets the critical circle (center $(1, 0)$, radius 1) if and only if $p \in \{2, 3\}$. Five theorems, zero errors, zero sorry.

Keywords: Lean, formalization, critical circle, prime circle, intersections, 0pat

1 The observation

The Riemann direction has four central geometric objects in the complex plane: the real axis, the imaginary axis, the critical circle (center $(1, 0)$, radius 1, the image of the critical line under reciprocal), and the prime circle of each prime p (center 0, radius $\sqrt{p}$, the locus of the two-square representations of p). This exercise observes their intersections — the complete table, every entry machine-checked.

Pair	Intersection	Status
real axis $\cap$ imaginary axis	$\{0\}$	definition
real axis $\cap$ critical circle	$\{0, 2\}$	C020 (proved)
imaginary axis $\cap$ critical circle	$\{0\}$	new
real axis $\cap$ prime circle p	$\{\pm\sqrt{p}\}$	new
imaginary axis $\cap$ prime circle p	$\{\pm i\sqrt{p}\}$	new
prime-2 circle $\cap$ critical circle	$\{1 \pm i\}$	C020 (proved)
prime-3 circle $\cap$ critical circle	$\{3/2 \pm \sqrt{3}/2 \cdot i\}$	new
prime circle $p \geq 5 \cap$ critical circle	$\emptyset$	new (structural)

2 The theorems

The imaginary axis meets the critical circle only at the origin (*imagAxis_inter_criticalCircle*). A point $t \cdot i$ of the imaginary axis has distance $\sqrt{1 + t^2}$ from $(1, 0)$; it lies on the critical circle exactly when $1 + t^2 = 1$, i.e. $t = 0$. The

contrast with the real axis is exact: the real axis meets the critical circle at the two points $\{0, 2\}$ (C020), the imaginary axis at one point $\{0\}$.

The axes meet the prime circle in square roots (*realAxis_inter_primeCircle*, *imagAxis_inter_primeCircle*).

A real point r has norm r^2 ; it lies on the prime circle of p exactly when $r^2 = p$, i.e. $r = \pm\sqrt{p}$. Similarly, the imaginary point $t \cdot i$ lies on the prime circle exactly when $t^2 = p$. Both intersections are irrational pairs for prime p ; they are never lattice points.

The prime-3 circle meets the critical circle (*prime3Circle_inter_criticalCircle*).

The point $\frac{3}{2} + \frac{\sqrt{3}}{2}i$ has norm $9/4 + 3/4 = 3$ and distance $\sqrt{1/4 + 3/4} = 1$ from $(1, 0)$. Together with the prime-2 intersection $\{1 \pm i\}$ (C020): the small primes 2 and 3 have circles that meet the critical circle.

The structural theorem: circles of primes $p \geq 5$ do not meet the critical circle (*primeCircle_inter_criticalCircle_ge5*).

Suppose $z = a + bi$ lies on both circles. Then $\|z\|^2 = p$ and $\|z - 1\|^2 = 1$. Subtracting gives $-2a + 1 = 1 - p$, so $a = p/2$; then $b^2 = p - a^2 = p - p^2/4 = p(4 - p)/4$, which is negative for $p \geq 5$ — contradicting $b^2 \geq 0$. Hence the prime circle of $p \geq 5$ is disjoint from the critical circle. Equivalently: the prime circle meets the critical circle if and only if $p \in \{2, 3\}$.

3 Honest boundary and position

The content is classical geometry (circle intersections in the plane); the value is observational and structural. The observation completes the intersection table of the Riemann direction's four objects and adds one new structural fact: only the prime circles of 2 and 3 touch the critical circle; all other prime circles are disjoint from it. No claim is made about the zeros of ζ : the critical circle is the geometric image of the critical line, and its intersections with the prime circles describe the geometry of the plane, not the analysis of the function.

4 Formalization

The proof is a single Lean file (*proof.lean*, 108 lines), importing *Formal.ZeroRelative.ComplexAxis* (the base construction of the Riemann chain, C011) and *Mathlib.Analysis.Real.Sqrt*. Five theorems, compiled with Lean 4.32.2 / *mathlib* v4.32.2, zero errors, zero sorry.

5 Conclusion

Theorem	Statement	Proof core
imagAxis_inter_criticalCircle	$t \cdot i \in \text{crit. circle} \leftrightarrow r = 0$	The four objects of the Riemann direction are now fully observed, their intersection table is complete and machine-checked, and the new structural fact stands — the prime circle touches the critical circle if and only if the prime is 2 or 3; for $p \geq 5$ the circles are disjoint. The geometry of the plane is now mapped exactly; the analysis that would connect it to the zeros of ζ remains the open span of the bridge.
realAxis_inter_primeCircle	$r \in \text{circle}_p \leftrightarrow r^2 = p$	
imagAxis_inter_primeCircle	$t \cdot i \in \text{circle}_p \leftrightarrow t = p$	
prime3Circle_inter_criticalCircle	$\frac{3}{2} + \frac{\sqrt{3}}{2}i$ in both	
primeCircle_inter_criticalCircle_ge5	$p \geq 5 \Rightarrow \text{disjoint}$	

References